DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK SECURITY (CYS360)
(6^{TH} SEMESTER, 3^{RD} Year) LAB EXPERIMENT # 14

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Score:	__________	Date:	____________	Instructor's Signature:	______________

Lab # 14: To configure IPSec VPN Between the Headquarters and Branch Offices (IP Address of the Headquarters Is Fixed)

Objective: The objective of this lab is to configure IPSec VPN between the HQ and Branch Offices. **Duration: 3 Hours**

Expected Outcomes: By the end of this lab session, students will be able to:

1. To configure an IPSec VPN tunnel between HQ and Branch.
2. To establish IPSec tunnels between FW_A–FW_B and FW_A–FW_C, ensuring that FW_B and FW_C cannot directly form an IPSec tunnel with each other.

Rubrics

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT
SCORE			

Note: Student performance will be assessed based on this rubric during lab sessions.

Networking Requirements

As shown in Figure 1, the headquarters is connected to two branches (Branch 1 and Branch 2). The networking requirements are as follows:

- FW_B connects Branch 1 to the Internet, and FW_C connects Branch 2 to the Internet.
- FW_A and FW_B are reachable to each other; and FW_A and FW_C are reachable to each other.
- FW_A and FW_B use fixed public IP addresses, and FW_C uses a dynamic public IP address.

The purposes of this networking are as follows:

- PC 2 and PC 3 in the branches can securely communicate with PC 1 in the headquarters. • IPSec tunnels can be set up between FW_A and FW_B, and between FW_A and FW_C. However, FW_B and FW_C cannot establish an IPSec tunnel with each other.

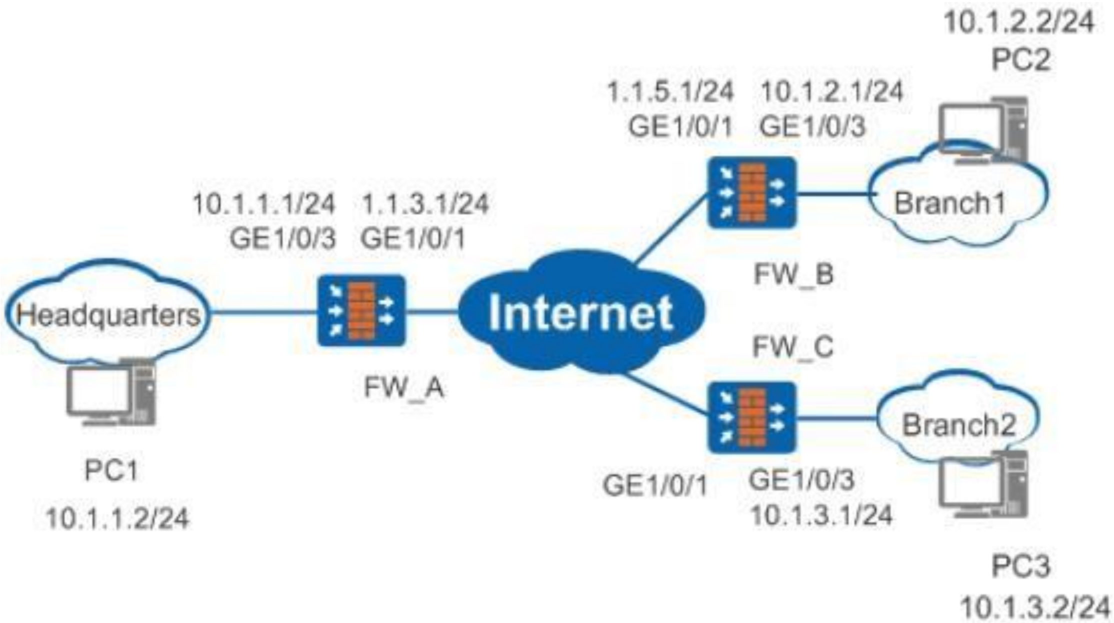


Figure 1 Networking diagram for configuring IPSEC Tunnel

Data Plan

Item	Data
FW_A	Interface number: GE1/0/3 IP address: 10.1.1.1/24
	Interface number: GE1/0/1 IP address: 1.1.3.1/24

	IPSec configuration Peer IP address: The IP address of FW_B is specified, but the IP address of FW_C is not specified. Authentication type: pre-shared key Pre-shared key: Test!1234 Local ID type: IP address Peer ID type: any
Item	**Data**
FW_B	Interface number: GE1/0/1 IP address: 1.1.5.1/24
	Interface number: GE1/0/3 IP address: 10.1.2.1/24
	IPSec configuration Peer IP address: 1.1.3.1 Authentication type: pre-shared key Pre-shared key: Test!1234 Local ID type: IP address Peer ID type: IP address
FW_C	Interface number: GE1/0/1 IP address: dynamic
	Interface number: GE1/0/3 IP address: 10.1.3.1/24

	IPSec configuration Peer IP address: 1.1.3.1 Authentication type: pre-shared key Pre-shared key: Test!1234 Local ID type: IP address Peer ID type: IP address

Configuration Roadmap

The roadmap for configuring FW_A, FW_B, and FW_C is similar:

1. Configure interfaces and routes, and enable security policies.
2. Configure the IPSec policy, including basic IPSec policy information, data flow to be protected by IPSec, and proposal parameters for security association (SA) negotiation.

Procedure

1. Perform basic configurations on FW_A, including setting the interface IP addresses, adding interfaces to security zones, and configuring interzone security policies and a static route.

a. Set interface IP addresses.

```
<sysname> system-view
[sysname] sysname FW_A
[FW_A] interface gigabitethernet 1/0/3
[FW_A-GigabitEthernet1/0/3] ip address 10.1.1.1 24
[FW_A-GigabitEthernet1/0/3] quit
[FW_A] interface gigabitethernet 1/0/1
[FW_A-GigabitEthernet1/0/1] ip address 1.1.3.1 24
[FW_A-GigabitEthernet1/0/1] quit
```

b. Add interfaces to corresponding security zones.

```
[FW_A] firewall zone trust
[FW_A-zone-trust] add interface gigabitethernet 1/0/3
[FW_A-zone-trust] quit
```

```
[FW_A] firewall zone untrust
[FW_A-zone-untrust] add interface gigabitethernet 1/0/1
[FW_A-zone-untrust] quit
```

c. Configure the security policies between the Trust and Untrust zones.

```
[FW_A] security-policy
[FW_A-policy-security] rule name policy1
[FW_A-policy-security-rule-policy1] source-zone trust
[FW_A-policy-security-rule-policy1] destination-zone untrust
[FW_A-policy-security-rule-policy1] source-address 10.1.1.0 24
[FW_A-policy-security-rule-policy1] destination-address 10.1.2.0 24
[FW_A-policy-security-rule-policy1] destination-address 10.1.3.0 24
[FW_A-policy-security-rule-policy1] action permit
[FW_A-policy-security-rule-policy1] quit
[FW_A-policy-security] rule name policy2
[FW_A-policy-security-rule-policy2] source-zone untrust
[FW_A-policy-security-rule-policy2] destination-zone trust
[FW_A-policy-security-rule-policy2] source-address 10.1.2.0 24
[FW_A-policy-security-rule-policy2] source-address 10.1.3.0 24
[FW_A-policy-security-rule-policy2] destination-address 10.1.1.0 24
[FW_A-policy-security-rule-policy2] action permit
[FW_A-policy-security-rule-policy2] quit
```

d. Configure the security policies between the Local and Untrust zones.

```
[FW_A-policy-security] rule name policy3
[FW_A-policy-security-rule-policy3] source-zone local
[FW_A-policy-security-rule-policy3] destination-zone untrust
[FW_A-policy-security-rule-policy3] source-address 1.1.3.1 32
[FW_A-policy-security-rule-policy3] action permit
[FW_A-policy-security-rule-policy3] quit
[FW_A-policy-security] rule name policy4
[FW_A-policy-security-rule-policy4] source-zone untrust
[FW_A-policy-security-rule-policy4] destination-zone local
[FW_A-policy-security-rule-policy4] destination-address 1.1.3.1 32
[FW_A-policy-security-rule-policy4] action permit
[FW_A-policy-security-rule-policy4] quit
[FW_A-policy-security] quit
```

e. Configure a static route to reach branches. Assume that the next hop of the route is 1.1.3.2.

```
[FW_A] ip route-static 0.0.0.0 0.0.0.0 1.1.3.2
```

3. Configure an IPSec policy and apply the policy to the corresponding interface on FW_A.

a. Configure ACLs to define data flows that need to be protected.

   i. Configure advanced ACL 3000 to define the data flow from the headquarters to Branch1.

```
[FW_A] acl 3000
[FW_A-acl-adv-3000] rule 5 permit ip source 10.1.1.0 0.0.0.255 destination 10.1.2.0 0.0.0.255
[FW_A-acl-adv-3000] rule 10 permit ip source 10.1.3.0 0.0.0.255 destination 10.1.2.0 0.0.0.255
[FW_A-acl-adv-3000] quit
```

   ii. Configure advanced ACL 3001 to define the data flow from the headquarters to Branch2.

```
[FW_A] acl 3001
[FW_A-acl-adv-3001] rule 5 permit ip source 10.1.1.0 0.0.0.255 destination 10.1.3.0 0.0.0.255
[FW_A-acl-adv-3001] rule 10 permit ip source 10.1.2.0 0.0.0.255 destination 10.1.3.0 0.0.0.255
[FW_A-acl-adv-3001] quit
```

b. Configure an IPSec proposal.

```
[FW_A] ipsec proposal tran1
[FW_A-ipsec-proposal-tran1] esp authentication-algorithm sha2-256
[FW_A-ipsec-proposal-tran1] esp encryption-algorithm aes-256
[FW_A-ipsec-proposal-tran1] quit
```

c. Configure an IKE proposal.

```
[FW_A] ike proposal 10
[FW_A-ike-proposal-10] authentication-method pre-share
[FW_A-ike-proposal-10] prf hmac-sha2-256
[FW_A-ike-proposal-10] encryption-algorithm aes-256
[FW_A-ike-proposal-10] dh group14
[FW_A-ike-proposal-10] integrity-algorithm hmac-sha2-256
[FW_A-ike-proposal-10] quit
```

d. Configure IKE peers.

   i. Configure an IKE peer named **b**.

```
[FW_A] ike peer b
[FW_A-ike-peer-b] ike-proposal 10
[FW_A-ike-peer-b] remote-address 1.1.5.1
[FW_A-ike-peer-b] pre-shared-key Test!1234
[FW_A-ike-peer-b] quit
```

ii. Configure an IKE peer named **c**.

```
[FW_A] ike peer c
[FW_A-ike-peer-c] ike-proposal 10
[FW_A-ike-peer-c] pre-shared-key Test!1234
[FW_A-ike-peer-c] quit
```

e. Configure an IPSec policy numbered **10** in IPSec policy group **map1**.

```
[FW_A] ipsec policy map1 10 isakmp
[FW_A-ipsec-policy-isakmp-map1-10] security acl 3000
[FW_A-ipsec-policy-isakmp-map1-10] proposal tran1
[FW_A-ipsec-policy-isakmp-map1-10] ike-peer b
[FW_A-ipsec-policy-isakmp-map1-10] quit
```

f. Configure an IPSec policy template with the name **map_temp** and number **1**.

```
[FW_A] ipsec policy-template map_temp 1
[FW_A-ipsec-policy-templet-map_temp-1] security acl 3001
[FW_A-ipsec-policy-templet-map_temp-1] proposal tran1
[FW_A-ipsec-policy-templet-map_temp-1] ike-peer c
[FW_A-ipsec-policy-templet-map_temp-1] quit
```

g. Refer to IPSec policy template **map_temp** in IPSec policy **20** of IPSec policy group **map1**.

```
[FW_A] ipsec policy map1 20 isakmp template map_temp
```

h. Apply IPSec policy group **map1** to GE1/0/1.

```
[FW_A] interface gigabitethernet 1/0/1
[FW_A-GigabitEthernet1/0/1] ipsec policy map1
[FW_A-GigabitEthernet1/0/1] quit
```

4. Perform basic configurations on FW_B.

a. Set interface IP addresses and add the interfaces to security zones.

i. Set interface IP addresses.

```
<sysname> system-view
[sysname] sysname FW_B
```

```
[FW_B] interface gigabitethernet 1/0/3
[FW_B-GigabitEthernet1/0/3] ip address 10.1.2.1 24
[FW_B-GigabitEthernet1/0/3] quit
[FW_B] interface gigabitethernet 1/0/1
[FW_B-GigabitEthernet1/0/1] ip address 1.1.5.1 24
[FW_B-GigabitEthernet1/0/1] quit
```

ii. Add GE1/0/3 to the Trust zone and GE1/0/1 to the Untrust zone.

```
[FW_B] firewall zone trust
[FW_B-zone-trust] add interface gigabitethernet 1/0/3
[FW_B-zone-trust] quit
[FW_B] firewall zone untrust
[FW_B-zone-untrust] add interface gigabitethernet 1/0/1
[FW_B-zone-untrust] quit
```

b. Configure interzone security policies.

i. Configure the security policies between the Trust and Untrust zones.

```
[FW_B] security-policy
[FW_B-policy-security] rule name policy1
[FW_B-policy-security-rule-policy1] source-zone trust
[FW_B-policy-security-rule-policy1] destination-zone untrust
[FW_B-policy-security-rule-policy1] source-address 10.1.2.0 24
[FW_B-policy-security-rule-policy1] destination-address 10.1.1.0 24
[FW_B-policy-security-rule-policy1] action permit
[FW_B-policy-security-rule-policy1] quit
[FW_B-policy-security] rule name policy2

[FW_B-policy-security-rule-policy2] source-zone untrust
[FW_B-policy-security-rule-policy2] destination-zone trust
[FW_B-policy-security-rule-policy2] source-address 10.1.1.0 24
[FW_B-policy-security-rule-policy2] destination-address 10.1.2.0 24
[FW_B-policy-security-rule-policy2] action permit
[FW_B-policy-security-rule-policy2] quit
```

ii. Configure the security policies between the Local and Untrust zones.

```
[FW_B-policy-security] rule name policy3
[FW_B-policy-security-rule-policy3] source-zone local
```

```
[FW_B-policy-security-rule-policy3] destination-zone untrust
[FW_B-policy-security-rule-policy3] source-address 1.1.5.1 32
[FW_B-policy-security-rule-policy3] destination-address 1.1.3.1 32
[FW_B-policy-security-rule-policy3] action permit
[FW_B-policy-security-rule-policy3] quit
[FW_B-policy-security] rule name policy4
[FW_B-policy-security-rule-policy4] source-zone untrust
[FW_B-policy-security-rule-policy4] destination-zone local
[FW_B-policy-security-rule-policy4] source-address 1.1.3.1 32
[FW_B-policy-security-rule-policy4] destination-address 1.1.5.1 32
[FW_B-policy-security-rule-policy4] action permit
[FW_B-policy-security-rule-policy4] quit
[FW_B-policy-security] quit
```

c. Configure a static route to the headquarters and the other branch. Assume that the next hop is 1.1.5.2.

```
[FW_B] ip route-static 0.0.0.0 0.0.0.0 1.1.5.2
```

5. Configure an IPSec policy and apply the policy to the corresponding interface on FW_B.

   a. Configure an ACL to define data flows that need to be protected.

```
[FW_B] acl 3000
[FW_B-acl-adv-3000] rule 5 permit ip source 10.1.2.0 0.0.0.255 destination
   10.1.1.0 0.0.0.255
[FW_B-acl-adv-3000] rule 10 permit ip source 10.1.2.0 0.0.0.255 destination
   10.1.3.0 0.0.0.255
[FW_B-acl-adv-3000] quit
```

   b. Configure an IPSec proposal.

```
[FW_B] ipsec proposal tran1
[FW_B-ipsec-proposal-tran1] esp authentication-algorithm sha2-256
[FW_B-ipsec-proposal-tran1] esp encryption-algorithm aes-256
[FW_B-ipsec-proposal-tran1] quit
```

   c. Configure an IKE proposal.

```
[FW_B] ike proposal 10
[FW_B-ike-proposal-10] authentication-method pre-share
[FW_B-ike-proposal-10] prf hmac-sha2-256
[FW_B-ike-proposal-10] encryption-algorithm aes-256
[FW_B-ike-proposal-10] dh group14
[FW_B-ike-proposal-10] integrity-algorithm hmac-sha2-256
[FW_B-ike-proposal-10] quit
```

d. Configure an IKE peer.

```
[FW_B] ike peer a
[FW_B-ike-peer-a] ike-proposal 10
[FW_B-ike-peer-a] remote-address 1.1.3.1
[FW_B-ike-peer-a] pre-shared-key Test!1234
[FW_B-ike-peer-a] quit
```

e. Configure an IPSec policy with the name **map1** and number **10**.

```
[FW_B] ipsec policy map1 10 isakmp
[FW_B-ipsec-policy-isakmp-map1-10] security acl 3000
[FW_B-ipsec-policy-isakmp-map1-10] proposal tran1
[FW_B-ipsec-policy-isakmp-map1-10] ike-peer a
[FW_B-ipsec-policy-isakmp-map1-10] quit
```

f. Apply security policy **map1** to GE1/0/1.

```
[FW_B] interface gigabitethernet 1/0/1
[FW_B-GigabitEthernet1/0/1] ipsec policy map1
[FW_B-GigabitEthernet1/0/1] quit
```

6. Perform basic configurations on FW_C.

a. Set interface IP addresses and add them to the corresponding security zones.

i. Set the IP address of GE1/0/3 to 10.1.3.1/24, and configure GE1/0/1 to obtain an IP address dynamically through DHCP.

```
<sysname> system-view
[sysname] sysname FW_C
[FW_C] interface gigabitethernet 1/0/3
[FW_C-GigabitEthernet1/0/3] ip address 10.1.2.1 24
[FW_C-GigabitEthernet1/0/3] quit
[FW_C] interface gigabitethernet 1/0/1
[FW_C-GigabitEthernet1/0/1] ip address dhcp-alloc
[FW_C-GigabitEthernet1/0/1] quit
```

ii. Add GE1/0/3 to the Trust zone and GE1/0/1 to the Untrust zone.

```
[FW_C] firewall zone trust
[FW_C-zone-trust] add interface gigabitethernet 1/0/3
[FW_C-zone-trust] quit
[FW_C] firewall zone untrust
[FW_C-zone-untrust] add interface gigabitethernet 1/0/1
[FW_C-zone-untrust] quit
```

b. Configure interzone security policies.

i. Configure the security policies between the Trust and Untrust zones.

```
[FW_C] security-policy
[FW_C-policy-security] rule name policy1
[FW_C-policy-security-rule-policy1] source-zone trust
[FW_C-policy-security-rule-policy1] destination-zone untrust
[FW_C-policy-security-rule-policy1] source-address 10.1.3.0 24
[FW_C-policy-security-rule-policy1] destination-address 10.1.1.0 24
[FW_C-policy-security-rule-policy1] action permit
[FW_C-policy-security-rule-policy1] quit
[FW_B-policy-security] rule name policy2
[FW_C-policy-security-rule-policy2] source-zone untrust
[FW_C-policy-security-rule-policy2] destination-zone trust
[FW_C-policy-security-rule-policy2] source-address 10.1.1.0 24
[FW_C-policy-security-rule-policy2] destination-address 10.1.3.0 24
[FW_C-policy-security-rule-policy2] action permit
[FW_C-policy-security-rule-policy2] quit
```

ii. Configure the security policies between the Local and Untrust zones.

```
[FW_C-policy-security] rule name policy3
[FW_C-policy-security-rule-policy3] source-zone local
[FW_C-policy-security-rule-policy3] destination-zone untrust
FW_C-policy-security-rule-policy3] destination-address 1.1.3.1 32
[FW_C-policy-security-rule-policy3] action permit
[FW_C-policy-security-rule-policy3] quit
[FW_C-policy-security] rule name policy4
[FW_C-policy-security-rule-policy4] source-zone untrust
[FW_C-policy-security-rule-policy4] destination-zone local
[FW_C-policy-security-rule-policy4] source-address 1.1.3.1 32
[FW_C-policy-security-rule-policy4] action permit
[FW_C-policy-security-rule-policy4] quit
[FW_C-policy-security] quit
```

c. Configure a static route to the headquarters and the other branch.

```
[FW_C] ip route-static 0.0.0.0 0.0.0.0 GigabitEthernet 1/0/1
```

7. Configure an IPSec policy and apply the policy to the corresponding interface on FW_C.

a. Configure an ACL to define data flows that need to be protected.

```
[FW_C] acl 3000
[FW_C-acl-adv-3000] rule 5 permit ip source 10.1.3.0 0.0.0.255 destination 10.1.1.0
0.0.0.255
[FW_C-acl-adv-3000] rule 10 permit ip source 10.1.3.0 0.0.0.255 destination 10.1.2.0
0.0.0.255
[FW_C-acl-adv-3000] quit
```

b. Configure an IPSec proposal.

```
[FW_C] ipsec proposal tran1
[FW_C-ipsec-proposal-tran1] esp authentication-algorithm sha2-256
[FW_C-ipsec-proposal-tran1] esp encryption-algorithm aes-256
[FW_C-ipsec-proposal-tran1] quit
```

c. Configure an IKE proposal.

```
[FW_C] ike proposal 10
[FW_C-ike-proposal-10] authentication-method pre-share
[FW_C-ike-proposal-10] prf hmac-sha2-256
[FW_C-ike-proposal-10] encryption-algorithm aes-256
[FW_C-ike-proposal-10] dh group14
[FW_C-ike-proposal-10] integrity-algorithm hmac-sha2-256
```

```
[FW_C-ike-proposal-10] quit
```

d. Configure an IKE peer.

```
[FW_C] ike peer a
[FW_C-ike-peer-a] ike-proposal 10
[FW_C-ike-peer-a] remote-address 1.1.3.1
[FW_C-ike-peer-a] pre-shared-key Test!1234
[FW_C-ike-peer-a] quit
```

8. Configure an IPSec policy with the name **map1** and number **10**.

```
[FW_C] ipsec policy map1 10 isakmp
[FW_C-ipsec-policy-isakmp-map1-10] security acl 3000
[FW_C-ipsec-policy-isakmp-map1-10] proposal tran1
[FW_C-ipsec-policy-isakmp-map1-10] ike-peer c
[FW_C-ipsec-policy-isakmp-map1-10] quit
```

9. Apply ISPec policy **map1** to GE1/0/1.

```
[FW_C] interface gigabitethernet 1/0/1
[FW_C-GigabitEthernet1/0/1] ipsec policy map1
[FW_C-GigabitEthernet1/0/1] quit
```

Verification

1. After the configuration is complete, PC2 and PC1 can access each other. They both can trigger the establishment of an IPSec SA. PC3 can proactively access PC1, and only FW_C of PC3 can trigger the establishment of an IPSec SA. After the IPSec SA is established, PC1 and PC3 can access each other. In this way, mutual access between PC2 and PC3 is implemented.

2. On FW_A, you can view two pairs of IKE SAs.

```
<FW_A> display ike sa
                                                                           Ike
                                                               sa information :
   Conn-ID        Peer            VPN             Flag(s)               Phase
  ----------------------------------------------------------------------------
   50336907       1.1.5.1                         RD|ST|A               v2:2
   50336906       1.1.5.1                         RD|ST|A               v2:1
   33554436       1.1.6.254                       RD|A                  v2:2
   33554435       1.1.6.254                       RD|A                  v2:1

  Number of SA entries  : 4
```

```
  Number of SA entries of all cpu : 4

                                                                    Flag Description:
  RD--READY   ST--STAYALIVE   RL--REPLACED   FD--FADING   TO--TIMEOUT
  HRT--HEARTBEAT   LKG--LAST KNOWN GOOD SEQ NO.   BCK--BACKED UP
M--ACTIVE   S--STANDBY   A--ALONE  NEG--NEGOTIATING
```

3. On FW_B and FW_C, you can view the IKE SAs whose peer ends are the headquarters. The following takes the information displayed on FW_B as an example.

```
<FW_B> display ike sa
                                                                                Ike
sa information :
Conn-ID       Peer            VPN            Flag(s)              Phase
  ------------------------------------------------------------------------------
    16782416      1.1.3.1                       RD|A                  v2:2
    16782415      1.1.3.1                       RD|A                  v2:1

  Number of SA entries  : 2

  Number of SA entries of all cpu : 2

                                                                    Flag Description:
  RD--READY   ST--STAYALIVE   RL--REPLACED   FD--FADING   TO--TIMEOUT
  HRT--HEARTBEAT   LKG--LAST KNOWN GOOD SEQ NO.   BCK--BACKED UP
M--ACTIVE   S--STANDBY   A--ALONE  NEG--NEGOTIATING
```

4. On FW_A, you can view two pairs of bidirectional IPSec SAs, corresponding to FW_B and FW_C respectively.

```
<FW_A> display ipsec sa brief

Current ipsec sa num:4
Number of SAs:4
    Src address   Dst address      SPI        VPN  Protocol     Algorithm
-------------------------------------------------------------------------------
    1.1.6.254      1.1.3.1       4001819557        ESP      E:AES-256 A:SHA2-256128
    1.1.5.1        1.1.3.1       3923280450        ESP      E:AES-256 A:SHA2-2-128
    1.1.3.1        1.1.6.254     4249128694        ESP      E:AES-256 A:SHA2-256128
    1.1.3.1        1.1.5.1       787858613         ESP      E:AES-256 A:SHA2-256128
```

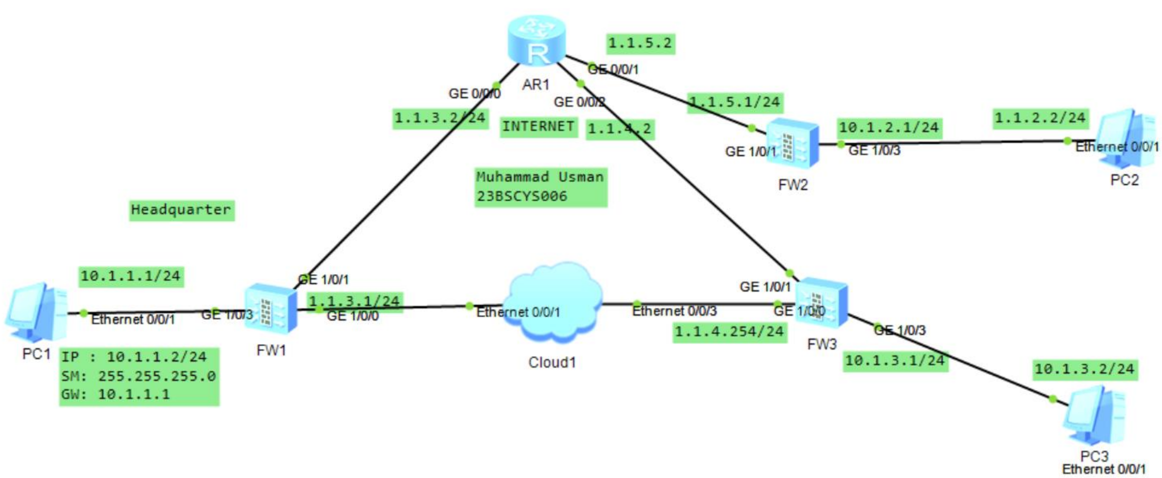
5. On FW_B and FW_C, you can view a pair of reverse IPSec SAs of FW_A. The following takes the information displayed on FW_B as an example.

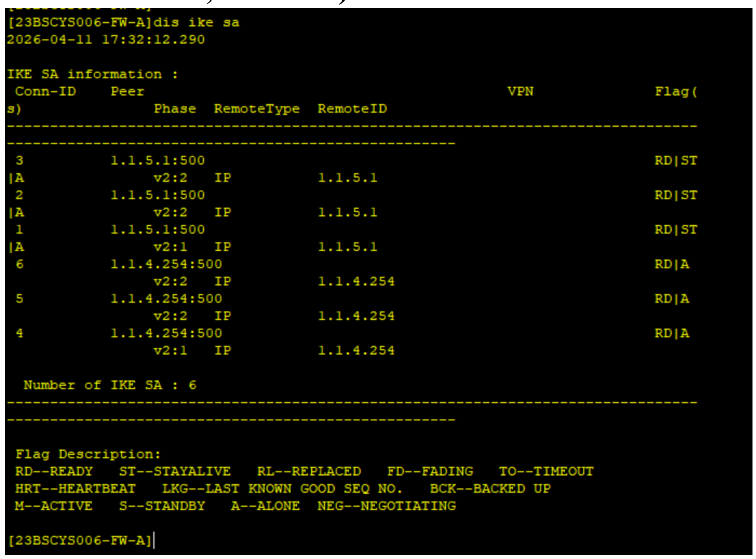
```
<FW_B> display ipsec sa brief
Current ipsec sa num:2
Number of SAs:2
    Src address    Dst address     SPI       VPN  Protocol     Algorithm
-------------------------------------------------------------------------------
     1.1.5.1         1.1.3.1     3923280450          ESP       E:AES-256 A:SHA2-256-128
     1.1.3.1         1.1.5.1     787858613           ESP       E:AES-256 A:SHA2-256-128
```

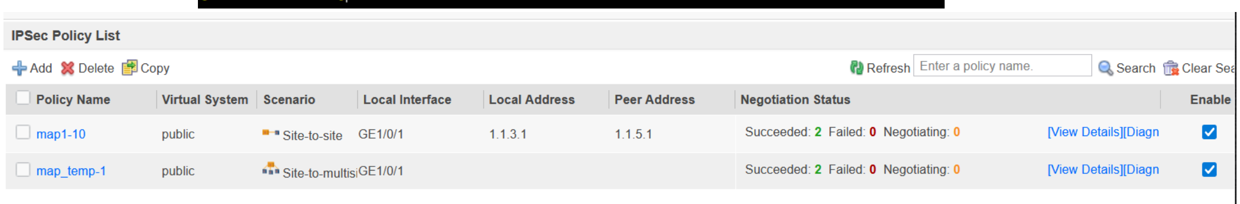
Task:

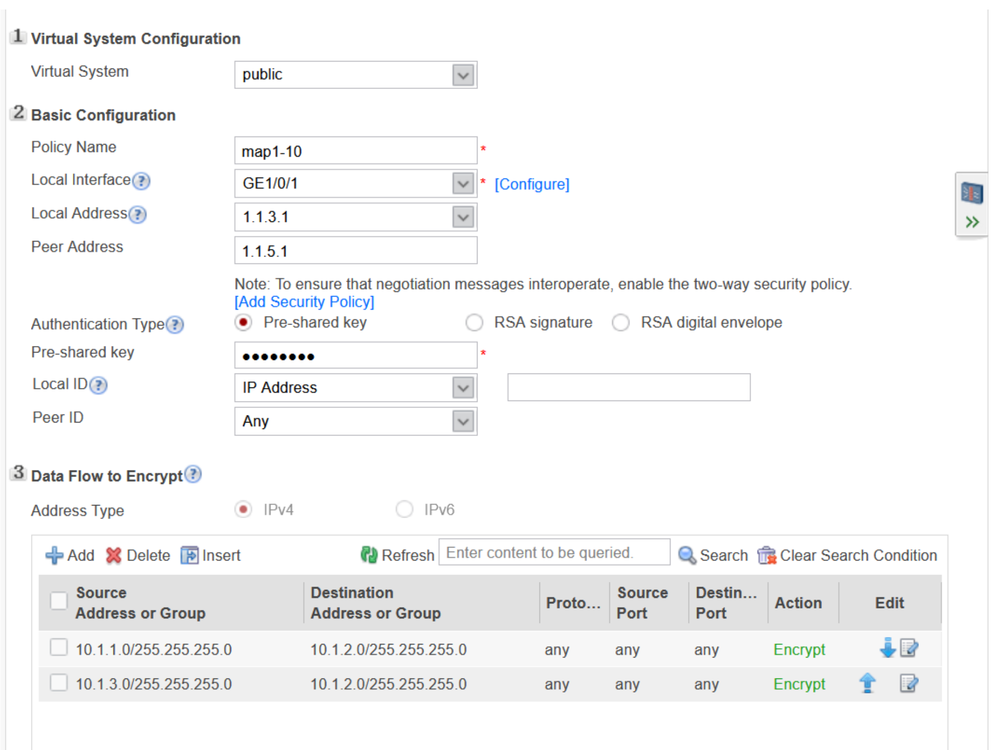
Attach Screenshots of all the topology and configuration.

Note: Attach all the screenshots of the configuration. The Firewall name should be your roll number like FW1-22BSCYS001. Upload the .topo and .cfg files on MS Teams Assignment.




[23BSCYS006-FW-A]dis ike sa
2026-04-11 17:32:12.290
IKE SA information :
Conn-ID Peer VPN Flag(
s) Phase RemoteType RemoteID
3 1.1.5.1:500 RD|ST
|A v2:2 IP 1.1.5.1
2 1.1.5.1:500 RD|ST
|A v2:2 IP 1.1.5.1
1 1.1.5.1:500 RD|ST
|A v2:1 IP 1.1.5.1
6 1.1.4.254:500 RD|A
v2:2 IP 1.1.4.254
5 1.1.4.254:500 RD|A
v2:2 IP 1.1.4.254
4 1.1.4.254:500 RD|A
v2:1 IP 1.1.4.254
Number of IKE SA : 6
Flag Description:
RD--READY ST--STAYALIVE RL--REPLACED FD--FADING TO--TIMEOUT
HRT--HEARTBEAT LKG--LAST KNOWN GOOD SEQ NO. BCK--BACKED UP
M--ACTIVE S--STANDBY A--ALONE NEG--NEGOTIATING
[23BSCYS006-FW-A]


IPSec Policy List
Add Delete Copy
Refresh Enter a policy name. Search Clear Sea
Policy Name Virtual System Scenario Local Interface Local Address Peer Address Negotiation Status Enable
map1-10 public Site-to-site GE1/0/1 1.1.3.1 1.1.5.1 Succeeded: 2 Failed: 0 Negotiating: 0 [View Details][Diagn
map_temp-1 public Site-to-multisi GE1/0/1 Succeeded: 2 Failed: 0 Negotiating: 0 [View Details][Diagn


1 Virtual System Configuration
Virtual System public
2 Basic Configuration
Policy Name map1-10
Local Interface GE1/0/1 [Configure]
Local Address 1.1.3.1
Peer Address 1.1.5.1
Note: To ensure that negotiation messages interoperate, enable the two-way security policy.
[Add Security Policy]
Authentication Type Pre-shared key RSA signature RSA digital envelope
Pre-shared key ••••••••
Local ID IP Address
Peer ID Any
3 Data Flow to Encrypt
Address Type IPv4 IPv6
Add Delete Insert Refresh Enter content to be queried. Search Clear Search Condition
Source Address or Group Destination Address or Group Proto... Source Port Destin... Port Action Edit
10.1.1.0/255.255.255.0 10.1.2.0/255.255.255.0 any any any Encrypt
10.1.3.0/255.255.255.0 10.1.2.0/255.255.255.0 any any any Encrypt

```
[23BSCYS006-FW-B]dis ike sa
2026-04-11 17:34:44.440

IKE SA information :
 Conn-ID    Peer                                         VPN              Flag(
s)               Phase  RemoteType  RemoteID
--------------------------------------------------------------------------------
----------------------------------------------------
 3          1.1.3.1:500                                                   RD|A
                 v2:2   IP          1.1.3.1
 2          1.1.3.1:500                                                   RD|A
                 v2:2   IP          1.1.3.1
 1          1.1.3.1:500                                                   RD|A
                 v2:1   IP          1.1.3.1

  Number of IKE SA : 3
--------------------------------------------------------------------------------
----------------------------------------------------

 Flag Description:
 RD--READY   ST--STAYALIVE   RL--REPLACED   FD--FADING   TO--TIMEOUT
 HRT--HEARTBEAT   LKG--LAST KNOWN GOOD SEQ NO.   BCK--BACKED UP
 M--ACTIVE   S--STANDBY   A--ALONE  NEG--NEGOTIATING

[23BSCYS006-FW-B]
```

```
[23BSCYS006-FW-B]dis ipsec sa
2026-04-11 17:35:01.780

ipsec sa information:

===============================
Interface: GigabitEthernet1/0/1
===============================

  -----------------------------
  IPSec policy name: "map1"
  Sequence number  : 10
  Acl group        : 3000
  Acl rule         : 5
  Mode             : ISAKMP
  -----------------------------
    Connection ID     : 2
    Encapsulation mode: Tunnel
    Holding time      : 0d 0h 13m 47s
    Tunnel local      : 1.1.5.1:500
    Tunnel remote     : 1.1.3.1:500
    Flow source       : 10.1.2.0/255.255.255.0 0/0-65535
    Flow destination  : 10.1.1.0/255.255.255.0 0/0-65535

    [Outbound ESP SAs]
      SPI: 184576609 (0xb006a61)
      Proposal: ESP-ENCRYPT-AES-256 ESP-AUTH-SHA2-256-128
      SA remaining key duration (kilobytes/sec): 10485760/2773
      Max sent sequence-number: 1
      UDP encapsulation used for NAT traversal: N
      SA encrypted packets (number/bytes): 0/0
```

```
      Anti-replay window size: 1024

  -----------------------------
  IPSec policy name: "map1"
  Sequence number  : 10
  Acl group        : 3000
  Acl rule         : 10
  Mode             : ISAKMP
  -----------------------------
    Connection ID     : 3
    Encapsulation mode: Tunnel
    Holding time      : 0d 0h 13m 44s
    Tunnel local      : 1.1.5.1:500
    Tunnel remote     : 1.1.3.1:500
    Flow source       : 10.1.2.0/255.255.255.0 0/0-65535
    Flow destination  : 10.1.3.0/255.255.255.0 0/0-65535

    [Outbound ESP SAs]
      SPI: 197007695 (0xbbe194f)
      Proposal: ESP-ENCRYPT-AES-256 ESP-AUTH-SHA2-256-128
      SA remaining key duration (kilobytes/sec): 10485760/2775
      Max sent sequence-number: 1
      UDP encapsulation used for NAT traversal: N
      SA encrypted packets (number/bytes): 0/0

    [Inbound ESP SAs]
      SPI: 187777091 (0xb314043)
      Proposal: ESP-ENCRYPT-AES-256 ESP-AUTH-SHA2-256-128
      SA remaining key duration (kilobytes/sec): 10485760/2775
      Max received sequence-number: 1
      UDP encapsulation used for NAT traversal: N
      SA decrypted packets (number/bytes): 0/0
      Anti-replay : Enable
      Anti-replay window size: 1024
[23BSCYS006-FW-B]
```

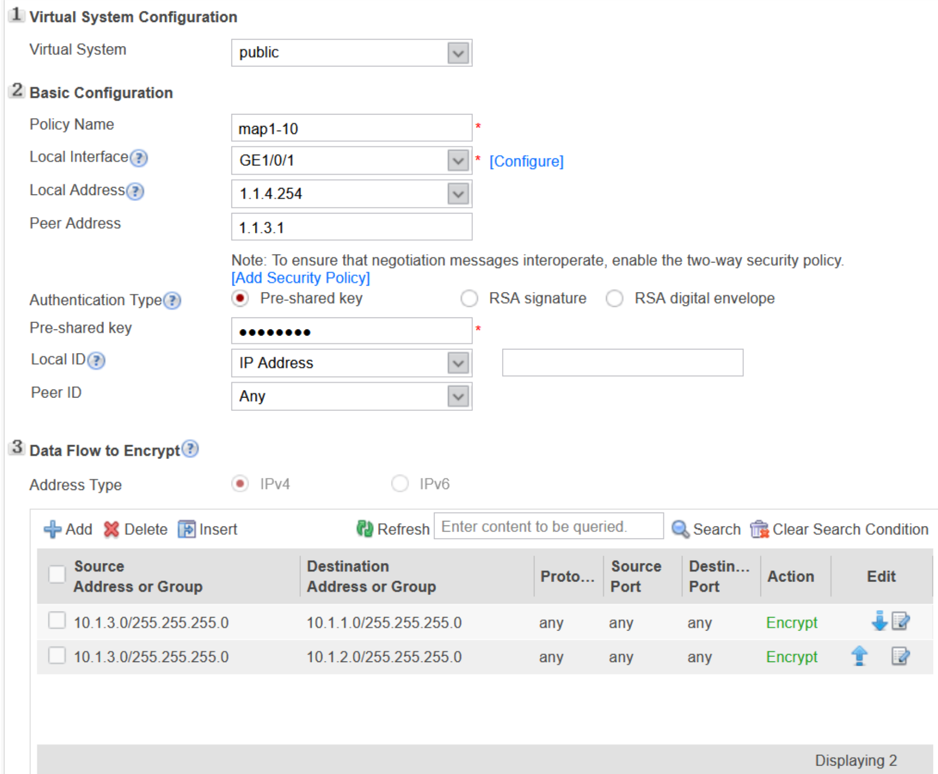
```
[23BSCYS006-FW-C]dis ike sa
2026-04-11 17:36:34.710

IKE SA information :
 Conn-ID    Peer                                        VPN              Flag(
s)              Phase  RemoteType  RemoteID
--------------------------------------------------------------------------------
----------------------------------------------------
 3          1.1.3.1:500                                                  RD|ST
|A              v2:2   IP          1.1.3.1
 2          1.1.3.1:500                                                  RD|ST
|A              v2:2   IP          1.1.3.1
 1          1.1.3.1:500                                                  RD|ST
|A              v2:1   IP          1.1.3.1

  Number of IKE SA : 3
--------------------------------------------------------------------------------
----------------------------------------------------

 Flag Description:
 RD--READY   ST--STAYALIVE   RL--REPLACED   FD--FADING   TO--TIMEOUT
 HRT--HEARTBEAT   LKG--LAST KNOWN GOOD SEQ NO.   BCK--BACKED UP
 M--ACTIVE   S--STANDBY   A--ALONE  NEG--NEGOTIATING

[23BSCYS006-FW-C]
```


1 Virtual System Configuration
Virtual System
public
2 Basic Configuration
Policy Name
map1-10
Local Interface
GE1/0/1
[Configure]
Local Address
1.1.4.254
Peer Address
1.1.3.1
Note: To ensure that negotiation messages interoperate, enable the two-way security policy.
[Add Security Policy]
Authentication Type
Pre-shared key
RSA signature
RSA digital envelope
Pre-shared key
Local ID
IP Address
Peer ID
Any
3 Data Flow to Encrypt
Address Type
IPv4
IPv6
Add
Delete
Insert
Refresh
Enter content to be queried.
Search
Clear Search Condition
Source Address or Group
Destination Address or Group
Proto...
Source Port
Destin... Port
Action
Edit
10.1.3.0/255.255.255.0
10.1.1.0/255.255.255.0
any
any
any
Encrypt
10.1.3.0/255.255.255.0
10.1.2.0/255.255.255.0
any
any
any
Encrypt
Displaying 2